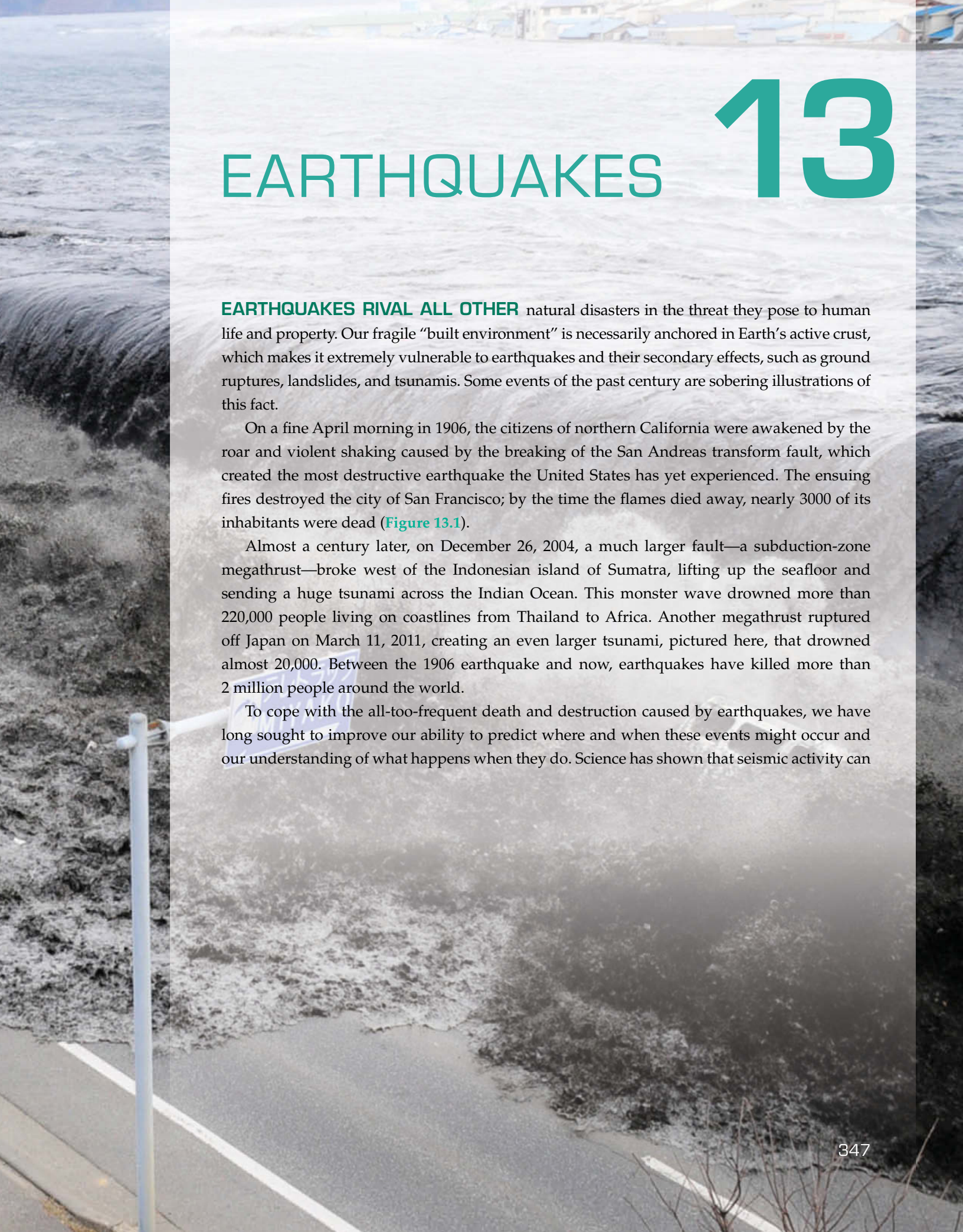




EARTHQUAKES

13

EARTHQUAKES RIVAL ALL OTHER natural disasters in the threat they pose to human life and property. Our fragile “built environment” is necessarily anchored in Earth’s active crust, which makes it extremely vulnerable to earthquakes and their secondary effects, such as ground ruptures, landslides, and tsunamis. Some events of the past century are sobering illustrations of this fact.

On a fine April morning in 1906, the citizens of northern California were awakened by the roar and violent shaking caused by the breaking of the San Andreas transform fault, which created the most destructive earthquake the United States has yet experienced. The ensuing fires destroyed the city of San Francisco; by the time the flames died away, nearly 3000 of its inhabitants were dead (**Figure 13.1**).

Almost a century later, on December 26, 2004, a much larger fault—a subduction-zone megathrust—broke west of the Indonesian island of Sumatra, lifting up the seafloor and sending a huge tsunami across the Indian Ocean. This monster wave drowned more than 220,000 people living on coastlines from Thailand to Africa. Another megathrust ruptured off Japan on March 11, 2011, creating an even larger tsunami, pictured here, that drowned almost 20,000. Between the 1906 earthquake and now, earthquakes have killed more than 2 million people around the world.

To cope with the all-too-frequent death and destruction caused by earthquakes, we have long sought to improve our ability to predict where and when these events might occur and our understanding of what happens when they do. Science has shown that seismic activity can



be understood in terms of the basic machinery of plate tectonics. As a result, attempts to reduce earthquake risk have become increasingly fused with the quest for a more fundamental understanding of the geologically active Earth.

This chapter will examine what happens during earthquakes, how scientists use seismic waves to locate and measure earthquakes, and what can be done to reduce the casualties and economic damage caused by earthquakes. We cannot yet reliably predict when large earthquakes will occur, but we can take measures to reduce their destructiveness. We can use our geologic knowledge to identify places where large earthquakes are likely; work with engineers to design buildings, dams, bridges, and other structures that can withstand seismic shaking; and help endangered communities prepare for and respond to seismic events.

■ What Is an Earthquake?

The movement of lithospheric plates generates enormous forces at the boundaries between those plates, as we saw in Chapter 7. These forces deform brittle crustal rocks in ways that can be described by the concepts of stress, strain, and strength. *Stress* is the local force per unit area that causes rocks to deform. *Strain* is the relative amount

of deformation, expressed as the percentage of distortion (for example, compression of a rock by 1 percent of its length). Rocks *fail*—that is, they lose cohesion and break into two or more parts—when they are stressed beyond a critical value, called their *strength*.

An **earthquake** occurs when rocks under stress suddenly fail along a geologic fault. Most large earthquakes are caused by ruptures of preexisting faults, where past earthquakes have already weakened the rocks on the



FIGURE 13.1 ■ This photograph, taken from a balloon by George Lawrence 5 weeks after the San Francisco earthquake of April 18, 1906, shows the devastation of the city caused by the quake and subsequent fires. The view is looking over Nob Hill toward the business district. [Corbis.]

fault surface. The two blocks of rock on either side of the fault slip suddenly, releasing energy in the form of seismic waves, which we feel as ground shaking. When the fault slips, the stress is reduced, dropping to a value below the rock strength. After the earthquake, the stress begins to increase again, eventually leading to another large earthquake. The faults involved in this repeated earthquake cycle are called *active faults*, and they are concentrated in the zones that form plate boundaries, where most of the stress and strain caused by plate movements is concentrated.

The Elastic Rebound Theory

The earthquake on the San Andreas fault that devastated San Francisco in 1906 was studied more than any earthquake had been up to that time. By mapping the displacement of the ground across the fault and analyzing seismic recordings of the earthquake, geologists showed that this rupture began just west of the Golden Gate and propagated along the fault strike for over 400 km, southeastward to the mission town of San Juan Bautista and northwestward all the way to Cape Mendocino (Figure 13.2). In 1910, one of the scientists who studied this rupture, Henry Fielding Reid of Johns Hopkins University, advanced the **elastic rebound theory** to explain how earthquakes recur on active faults in Earth's crust.

Picture a strike-slip fault between two crustal blocks, and imagine that surveyors have painted straight lines on the ground, running perpendicular to the fault and extending from one block to the other, as in Figure 13.3a. The two blocks are being pushed in opposite directions by plate movements. The weight of the overlying rock

presses them together, however, so friction locks them in place along the fault. They do not move, just as a pushed car does not move when the emergency brake is engaged. Instead of slipping along the fault as stress builds up, the blocks are strained elastically near the fault, as shown by the bent lines in Figure 13.3b. By *elastically*, we mean that the blocks would spring back and return to their undeformed, stress-free shape if the fault were suddenly to unlock.

As the slow plate movements continue to push the blocks in opposite directions, the elastic strain in the rocks—measured by the bending of the survey lines—continues to build up over decades, centuries, or even millennia (Figure 13.3c). At some point, the strength of the rocks is exceeded. Somewhere along the fault surface, the frictional bond that locks the fault can no longer hold, and it breaks. The blocks slip suddenly in a rupture that starts at a point but quickly extends over a section of the fault.

Figure 13.3d shows how the two blocks have rebounded—sprung back to their undeformed state—after the earthquake. The bent survey lines have straightened, and the two blocks have been displaced. (Note that a fence built just before the rupture has been bent during the rebound.) The distance of the displacement is called the **fault slip**. In the inset photograph of Figure 13.3, you can see that the fault slip during the 1906 earthquake was about 4 m. The peak velocity of the slipping at any point on the fault is about 1 m/s, so the entire episode of fault slipping at a given point takes only a few seconds. After the fault has slipped, it locks up again. The steady movement of the blocks on either side of the fault then causes stress to rise again, and the earthquake cycle repeats.

The energy that is slowly built up by elastic strain as the two blocks are pushed in opposite directions is like the elastic energy stored in a rubber band when it is slowly stretched. The sudden release of energy when a fault slips is like the violent backlash, or *rebound*, that occurs when the rubber band breaks. Some of this elastic energy is radiated in seismic waves, which can cause violent shaking many kilometers away from the fault.

The elastic rebound theory implies that there should be a periodic buildup and release of elastic energy at faults, and that the time between ruptures, called the **recurrence interval**, should be a constant (as shown in the bottom panel of Figure 13.3). The recurrence interval can be calculated by dividing the fault slip in each rupture by the long-term slip rate. The long-term slip rate on the San Andreas fault is about 30 mm/year, for example, so earthquakes with 4 m of slip should occur with a recurrence interval of about once every 130 years.

However, most active faults, including the San Andreas, do not conform to this simple theory. For instance, all of the strain accumulated since the last earthquake may not be released in the next—that is, the rebound may be incomplete—or the stress on one fault may change because

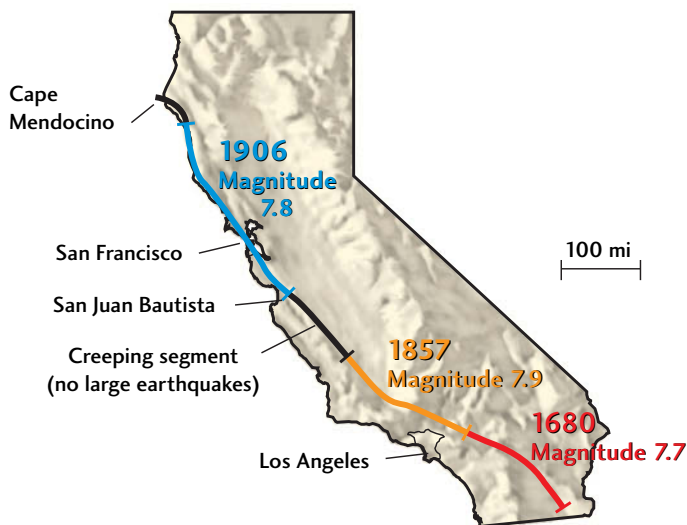


FIGURE 13.2 ■ Map of California, showing the segments of the San Andreas fault that ruptured in 1680, 1857, and 1906. [Southern California Earthquake Center.]

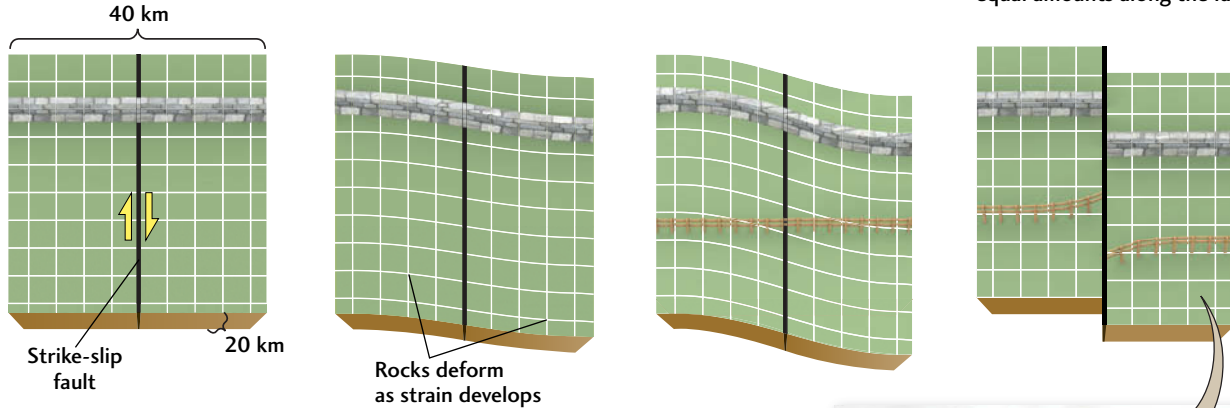
ROCKS DEFORM ELASTICALLY, THEN REBOUND DURING AN EARTHQUAKE RUPTURE

A A farmer builds a stone wall across a right-lateral strike-slip fault a few years after its last rupture.

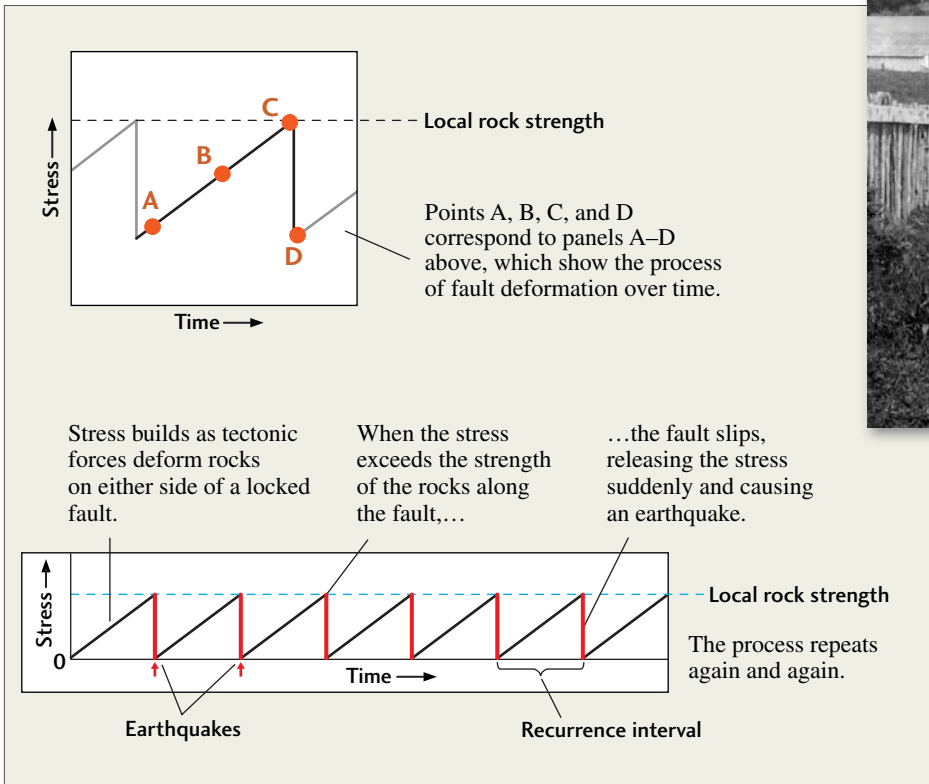
B Over the next 150 years, the relative motion of the blocks on either side of the fault causes the ground and the stone wall to deform.

C Just before the next rupture, a new fence is built across the already deformed land.

D The fault slips, lowering the stress, and the elastic rebound restores the blocks to their prestressed state. Both the rock wall and the fence are shifted equal amounts along the fault.



STRESS BUILDS UNTIL IT EXCEEDS ROCK STRENGTH



A fence built across the San Andreas fault near Bolinas, California, is offset by nearly 4 m after the great San Francisco earthquake of 1906.

FIGURE 13.3 ■ The elastic rebound theory explains the earthquake cycle. According to the theory, stress on rocks builds up over time as a result of plate movements. Earthquakes occur when that stress exceeds rock strength. Rocks under stress deform elastically, then rebound when an earthquake occurs. Panels A–D show deformation at the points labeled A–D in the bottom panel. [G. K. Gilbert/USGS.]

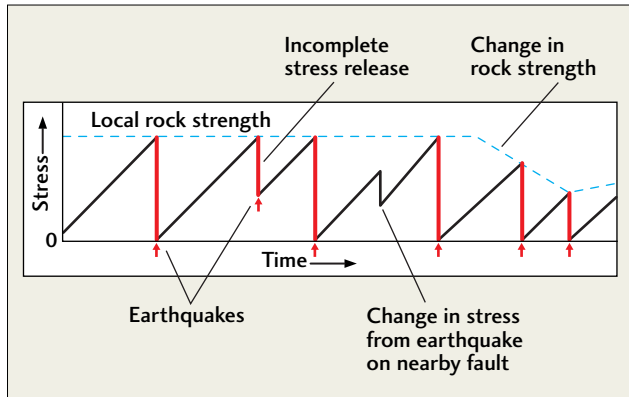


FIGURE 13.4 ■ Irregularities in the earthquake cycle can be caused by incomplete stress release, changes in stress caused by earthquakes on nearby faults, and local variations in rock strength.

of earthquakes on nearby faults (Figure 13.4). Over the long term, the strength of the fault rocks themselves may change. These irregularities are some of the reasons why earthquakes are so difficult to predict.

Fault Rupture During Earthquakes

The point at which fault slipping begins is the **focus** of an earthquake (Figure 13.5). The **epicenter** is the geographic point on Earth's surface directly above the focus. For example, you might hear in a news report, "The U.S. Geological Survey reports that the epicenter of last night's destructive earthquake in California was located 6 kilometers east of Los Angeles City Hall. The depth of the focus was 10 kilometers."

The focal depths of most earthquakes in continental crust range from 2 to 20 km. Continental earthquakes below 20 km are rare because under the high temperatures and pressures found at those greater depths, continental crust behaves as a ductile rather than a brittle material (just as hot wax flows when stressed, whereas cold wax breaks; see Chapter 7). In subduction zones, however, where cold oceanic lithosphere plunges into the mantle, earthquakes can originate at depths of almost 700 km.

The fault rupture does not happen all at once. It begins at the focus and expands outward along the fault surface, typically at 2 to 3 km/s (see Figure 13.5). The rupture stops where the stresses become insufficient to continue breaking the fault (where the rocks are stronger) or where the rupture enters ductile material in which it can no longer propagate. As we will see later in this chapter, the magnitude of an earthquake is related to the total area of fault rupture. Most earthquakes are very small, with rupture dimensions much less than the depth of the focus, so that the rupture never breaks the ground surface.

In large, destructive earthquakes, however, surface breaks are common. The 1906 San Francisco earthquake caused surface displacements averaging about 4 m along the 470-km section of the San Andreas fault that ruptured in that event (see the inset of Figure 13.3). Fault ruptures in the largest earthquakes can extend for more than 1000 km, and the fault slip can be tens of meters. Generally, the longer the fault rupture, the greater the fault slip.

As we have seen, the sudden slipping of the blocks at the time of the earthquake reduces the stress on the fault and releases much of the stored elastic energy. Most of this stored energy is converted to frictional heat in the fault zone or dissipated by rock fracturing, but part of it is released as seismic waves that travel outward from the rupture, much as waves ripple outward from the spot where a stone is dropped into a still pond. The focus of the earthquake generates the first seismic waves, but slipping parts of the fault continue to generate waves until the rupture stops. In a large event, the propagating rupture continues to produce waves for many tens of seconds. These waves can cause destruction in regions all along the fault rupture, even far from the epicenter. Towns along the San Andreas fault far north of San Francisco were badly damaged in the 1906 earthquake.

Foreshocks and Aftershocks

Almost all large earthquakes trigger smaller earthquakes called **aftershocks**. Aftershocks follow the triggering event, or **mainshock**, in sequences, and their foci are distributed in and around the rupture surface of the mainshock (Figure 13.6). Aftershock sequences exemplify the complexities of earthquakes that cannot be described by simple elastic rebound theory. Although fault slipping during the mainshock decreases the stress along most of the rupture surface, it can increase the stress on parts of the fault surface that did not slip, or where the slip was incomplete, as well as in surrounding regions. Aftershocks happen where that stress exceeds the rock strength.

The number and sizes of aftershocks depend on the magnitude of the mainshock, and their frequencies decrease with time after the mainshock. The Aftershocks of a magnitude 5 earthquake might last for only a few weeks, whereas those of a magnitude 7 earthquake occur over a larger region and can continue for several years. The size of the largest aftershock is generally about one magnitude unit smaller than the mainshock. According to this rough rule of thumb, a magnitude 7 earthquake might have an aftershock as large as magnitude 6.

In populated regions, the shaking from large aftershocks can be very dangerous, compounding the damage caused by the mainshock. On September 4, 2010, a magnitude 7.1 earthquake west of Christchurch, the second largest city in New Zealand, caused extensive damage, but nobody was killed and only a few people were injured. However, a magnitude 6.3 aftershock struck right beneath the center of Christchurch on February 22, 2011, collapsing

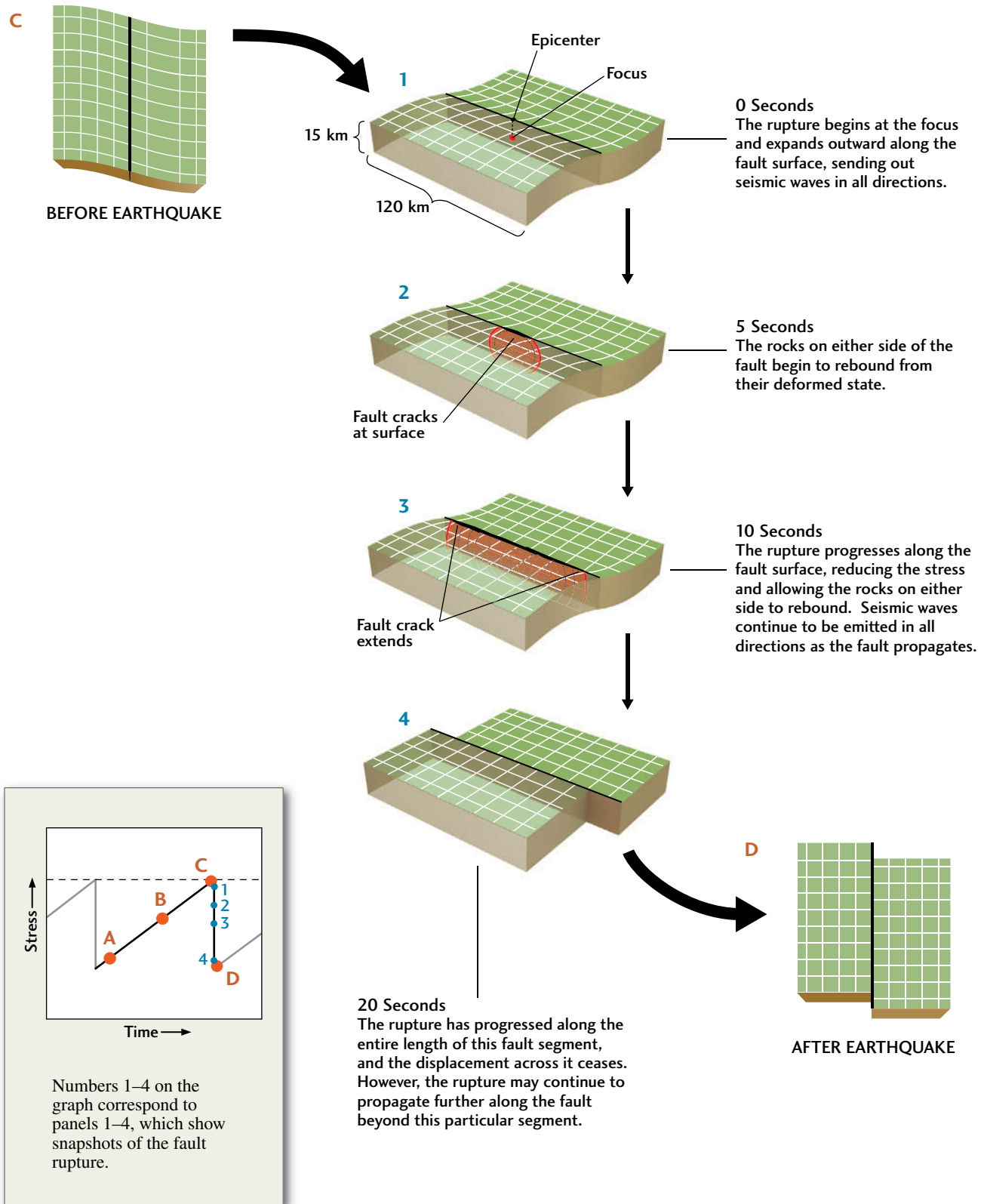


FIGURE 13.5 ■ During an earthquake, fault slipping begins at the focus and spreads out along the fault surface. Panels 1–4 are snapshots of the fault rupture corresponding to the numbered points on the graph.

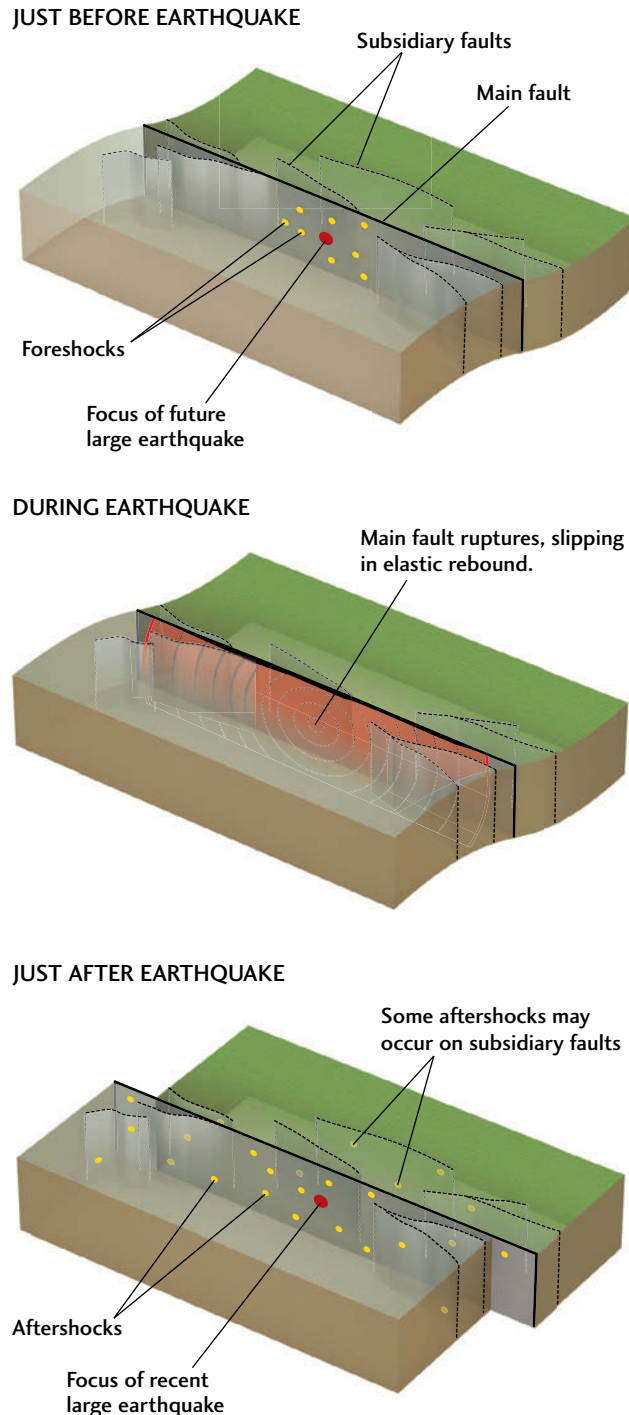


FIGURE 13.6 ■ Aftershocks are smaller earthquakes that follow a large earthquake (the mainshock). Foreshocks occur shortly before the mainshock, near its focus.

buildings and killing 185 people (Figure 13.7). The economic losses from this aftershock, estimated at \$15 billion, were several times greater than the losses caused by the mainshock five months before. Other strong aftershocks hit the city on June 13 and December, 2011, injuring dozens and causing \$4 billion in additional damages. More aftershocks can be expected in the years to come.



FIGURE 13.7 ■ Ruins of the Christchurch Catholic Cathedral, one of many buildings in downtown Christchurch, New Zealand, destroyed by an earthquake on February 22, 2011. This event was an aftershock of a larger but less damaging earthquake that occurred west of Christchurch on September 4, 2010. [EPA/DAVID WETHEY/Landov.]

A **foreshock** is a small earthquake that occurs shortly before a mainshock, near its focus (see Figure 13.6). One or more foreshocks have preceded many large earthquakes, so scientists have tried to use foreshocks to predict when and where large earthquakes might happen. Unfortunately, it is usually very hard to distinguish foreshocks from other small earthquakes that occur randomly and frequently on active faults, so this method has only rarely proved successful. The magnitude-9 Tohoku earthquake that caused a great tsunami to hit Honshu, Japan on March 11, 2011 (see Earth Issues 13.1), was preceded by a foreshock of magnitude 7.2 about 50 hours before the mainshock. In some sense, the mainshock was an anomalously big “aftershock” associated with the first event. But what turned out to be a foreshock was considered at the time to be only an ordinary magnitude-7.2 earthquake in the subduction zone.

As this example illustrates, foreshocks, mainshocks, and aftershocks can be classified in a definitive way only after the earthquake sequence has ended; during the sequence we cannot be sure whether the mainshock—the biggest event in the sequence—is yet to come.

■ How Do We Study Earthquakes?

As in any experimental science, instruments and field observations provide the basic data used to study earthquakes. These data enable investigators to locate earthquakes, determine their sizes and numbers, and understand their relationships to faults.

Seismographs

The **seismograph**, an instrument that records the seismic waves, is to the Earth scientist what the telescope is to the astronomer: a tool for peering into inaccessible regions (Figure 13.8). The ideal seismograph would be a device affixed to a stationary frame that is not attached to Earth. When the ground shook, the seismograph would measure the changing distance between the frame, which did not move, and the vibrating ground, which did. As yet, we have no way to position a seismograph that is not attached to Earth—although Global Positioning System (GPS) technology is beginning to remove this limitation. So we compromise. We attach a dense mass, such as a piece of steel, to Earth so loosely that the ground can vibrate up and down or side to side without causing much movement of the mass.

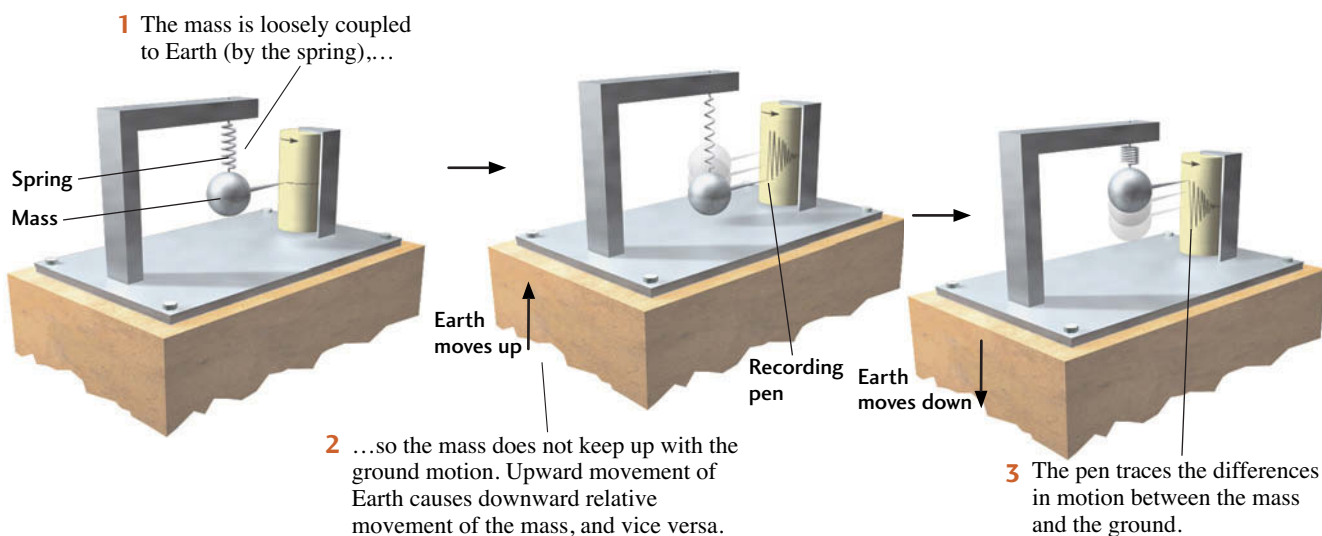
This loose attachment can be achieved by suspending the mass from a spring (Figure 13.8a). When seismic waves

move the ground up and down, the mass tends to remain stationary because of its inertia (an object at rest tends to stay at rest), but the mass and the ground move relative to each other because the spring can compress or stretch. In this way, the vertical displacement of the ground caused by seismic waves can be recorded by a pen on chart paper or, almost always these days, digitally on a computer. Such a record is called a *seismogram*.

Loose attachment of the mass can also be achieved with a hinge. A seismograph that has its mass suspended on hinges, like a swinging gate (Figure 13.8b), can record the horizontal displacement of the ground.

A typical seismographic station has seismographs set up to measure three components of ground movement: vertical, horizontal east-west, and horizontal north-south. Modern seismographs can detect ground oscillations of less than a billionth of a meter—an astounding feat, considering that such small displacements are of atomic size!

(a) Seismograph designed to detect vertical movement



(b) Seismograph designed to detect horizontal movement

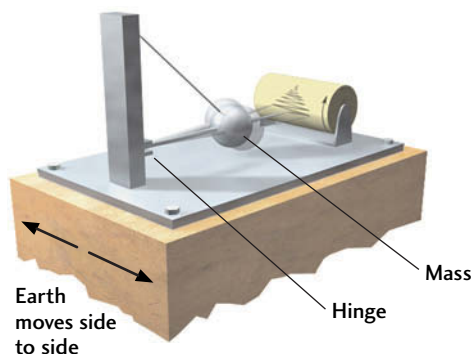


FIGURE 13.8 ■ A seismograph consists of a dense mass (such as a steel ball) attached to a recording device. Because of its inertia and its loose coupling to Earth through (a) a spring or (b) a hinge, the mass does not keep up with the motion of the ground.

Earth Issues

13.1 The Tsunami Stone of Aneyoshi

On a hill along the Tohoku coastline of northeastern Honshu, in the fishing hamlet of Aneyoshi, sits a stone monument of uncertain age, inscribed with Japanese characters that read, “High dwellings are the peace and harmony of our descendants. Remember the calamity of the great tsunamis. Do not build any homes below this point.” Aneyoshi, now part of the city of Miyako, was once more conveniently located down by the sea where the fishermen tied their boats, but only four of its residents survived the tsunami of 1896 and only two survived the tsunami of 1933. The stone reminds people why they now live on higher ground.

History became prophecy at 2:46 P.M. on March 11, 2011, when the offshore thrust fault that separates Japan from the Pacific plate began to slip. The rupture started on a small patch of the fault surface 30 kilometers beneath the ocean, about 100 km southeast of Aneyoshi, and accelerated outward like a crack through glass, reaching speeds of nearly 3 km/s (more than 6000 miles/hour). By the time it stopped several minutes later, the Pacific Plate had moved under Japan by as much as 40 m along a fault surface the size of South Carolina. Seismic waves from this Tohoku megaquake, which measured magnitude 9, propagated over Earth’s surface and through its deep interior, causing the planet to ring like a bell for many days.

The thrusting of Honshu eastward and upward over the Pacific plate raised the seafloor as much as 10 m almost instantly, displacing several hundred billion tons of water, which flowed away from the fault in a huge tsunami. In less than an hour, the water waves, slower than the seismic waves but much more deadly, passed into the bays and inlets of the Japanese coastline like an undulating monster, gaining height as they approached the shore. Funneled into harbors, the waves created immense walls of water—tsunami is Japanese for “harbor wave”—which inundated the near-shore communities, sweeping up boats, cars, and buildings, in some places travelling several kilometers inland.

The fast-moving swath of devastation was captured on horrific videos from helicopters overhead and by survivors

who made it to high ground and the tops of buildings. The tsunami overran the seawalls designed to protect the city center of Miyako, destroying all but 30 of the 1,000 boats in its famous fishing fleet and killing many hundreds who could not, or did not, escape in time. Though the exact number remains uncertain, the death toll along the Tohoku coastline was almost 20,000. One of the highest levels reached by the enormous wave—the greatest in recent Japanese history—was 39 m (128 ft) above the shoreline, just below the Aneyoshi tsunami stone. The residents in their houses above the stone were safe.



The tsunami stone at Aneyoshi. [Ko Sasaki/The New York Times/Redux.]

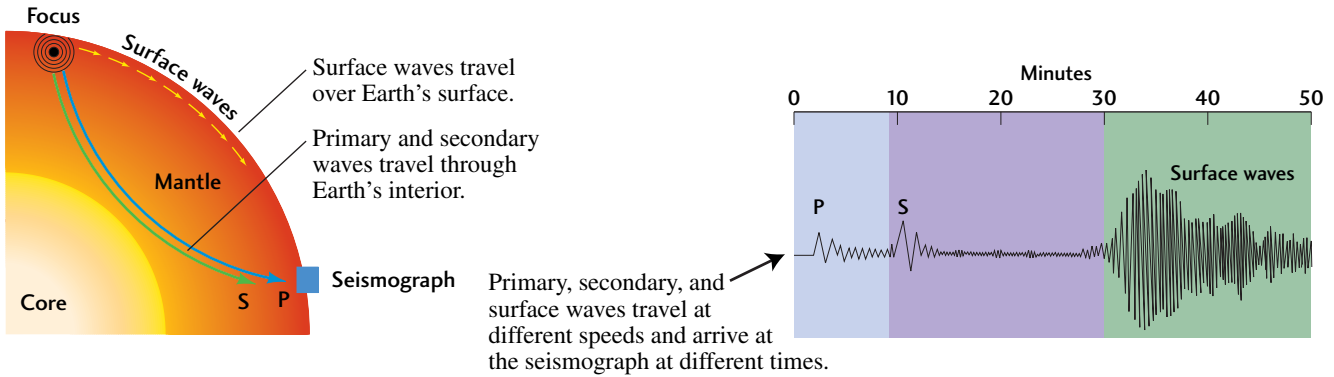
Seismic Waves

Install a seismograph almost anywhere, and within a few hours it will record the passage of seismic waves generated by an earthquake somewhere on Earth. The waves travel from the earthquake focus through Earth and arrive at the seismograph in three distinct groups (**Figure 13.9a**). The first waves to arrive are called primary waves, or **P waves**. The secondary waves, or **S waves**, follow. Both P waves

and S waves travel through Earth’s interior. Afterwards come the slower **surface waves**, which travel around Earth’s surface.

P waves in rock are similar to sound waves in air, except that P waves travel through the solid rock of Earth’s crust at about 6 km/s, which is about 20 times faster than sound waves travel through air. Like sound waves, P waves are *compressional waves*, so called because they travel through solid, liquid, or gaseous materials as a succession

(a) Seismic waves generated at an earthquake focus arrive at a seismograph far from the earthquake.

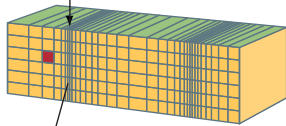


(b) Seismic waves are characterized by distinct types of ground deformation.

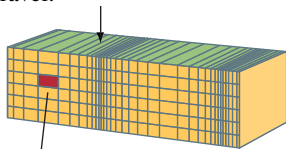
P-wave motion

P waves (primary waves) are compressional waves that travel quickly through rock.

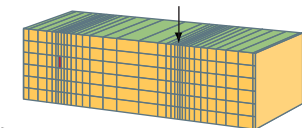
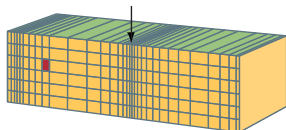
Compressional-wave crest



P waves travel as a series of contractions and expansions, pushing and pulling particles in the direction of their path of travel.



The red square charts the contraction and expansion of a section of rock.

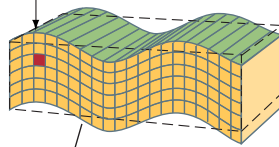


Wave direction

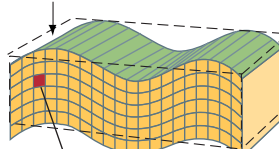
S-wave motion

S waves (secondary waves) travel at about half the speed of P waves.

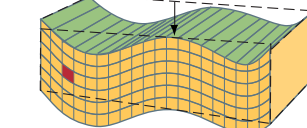
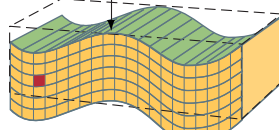
Shear-wave crest



S waves are shear waves that push material at right angles to their path of travel.



The red square shows how a section of rock shears from a square to a parallelogram as the S wave passes.

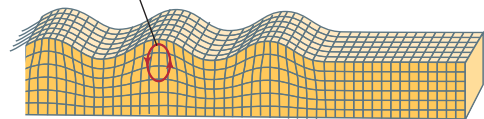


Wave direction

Surface-wave motion

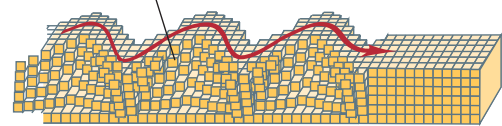
Surface waves ripple across Earth's surface, where air above the surface allows free movement. There are two types of surface waves.

In one type, the ground surface moves in a rolling, elliptical motion that decreases with depth beneath the surface.



Wave direction

In the second type, the ground shakes sideways, with no vertical motion.



Wave direction

FIGURE 13.9 (a) The three types of waves travel by different routes and at different speeds to a seismograph that records them. (b) The three types of seismic waves are characterized by distinct types of ground deformation. The red squares show the distortion of a section of rock as a wave passes through it.

of compressions and expansions (Figure 13.9b). P waves can be thought of as push-pull waves: they push or pull particles of matter in the direction of their path of travel.

S waves travel through solid rock at a little more than half the velocity of P waves. They are *shear waves* that displace material at right angles to their path of travel (see Figure 13.9b). Shear waves cannot travel through liquids or gases.

The velocities at which P and S waves travel are higher when the resistance to their movement is greater. It takes more force to compress solids than to shear them, so P waves always travel faster than S waves through a solid, which is why the P waves from an earthquake arrive at a seismograph before the S waves. This physical principle also explains why S waves cannot travel through air, water, or Earth's liquid outer core: gases and liquids put up no resistance to shear.

Surface waves are confined to Earth's surface and outer layers, like waves on the ocean. Their velocity is slightly less than that of S waves. One type of surface wave sets up a rolling motion in the ground; another type shakes the ground sideways (see Figure 13.9b). Surface waves are usually the most destructive waves in a large, shallow-focus earthquake, especially in sedimentary basins, where reverberations in the soft near-surface sediments can increase their amplitudes, causing much stronger shaking than in the basement rocks.

People have felt seismic waves and witnessed their destructiveness throughout history, but not until the close of the nineteenth century were scientists able to devise seismographs to record them accurately. Seismic waves enable us to study earthquakes but also provide our most important means of probing Earth's deep interior, as we will see in Chapter 14.

Locating the Focus

Locating an earthquake's focus is like deducing the distance of a lightning strike from the time interval between the flash of light and the sound of thunder: the greater the distance to the lightning bolt, the longer the time interval. Light travels faster than sound, so the lightning flash may be likened to the P waves of an earthquake and the thunder to the slower S waves.

The time interval between the arrival of P waves and S waves depends on the distance the waves have traveled from the focus: the longer the interval, the longer the distance the waves have traveled. Seismologists have used networks of sensitive seismographs around the world and highly accurate clocks to time the arrival of seismic waves from earthquakes as well as from underground nuclear test sites at known locations. From the results, they have constructed *travel-time curves*, which show how long it takes seismic waves of each type to travel a certain distance.

To estimate the distance to a new quake's focus, seismologists read from a seismogram the amount of time

between the arrival of the first P waves and the arrival of the first S waves. Then they use travel-time curves, like the ones shown in Figure 13.10, to determine the distance from the seismograph to the focus. If they can determine the distances from three or more seismographic stations, they can locate the focus (see Figure 13.10). They can also deduce the time of the quake at the focus—the earthquake's *origin time*—because the arrival time of the first P waves at each station is known, and it is possible to determine from the travel-time curves how long those waves took to reach the station. Today, this entire process is done automatically by computers, which use data from a large network of seismographs to determine each earthquake's epicenter, depth of focus, and origin time.

Measuring the Size of an Earthquake

Locating earthquakes is only one step on the way to understanding them. We must also determine their sizes, or *magnitudes*. Other things being equal (such as distance from the focus and regional geology), an earthquake's magnitude is the main factor that determines the intensity of the seismic waves it produces, and thus the earthquake's potential destructiveness.

RICHTER MAGNITUDE In 1935, Charles Richter, a California seismologist, devised a simple procedure that assigned a numerical size to each earthquake, now called the *Richter magnitude* (Figure 13.11). Richter studied astronomy as a young man and learned how astronomers use a logarithmic scale to measure the brightnesses of stars, which vary over a huge range of values. Adapting this idea to earthquakes, Richter took as his measure of earthquake size the logarithm of the largest ground movement registered by a standard type of seismograph at a standard distance, thus defining a **magnitude scale**.

On Richter's magnitude scale, two earthquakes at the same distance from a seismograph differ by one magnitude unit if the peak amplitude of the ground movement they produce differs by a factor of 10. The ground movement of an earthquake of magnitude 3, therefore, is 10 times that of an earthquake of magnitude 2. Similarly, a magnitude 6 earthquake produces ground movements that are 100 times greater than those of a magnitude 4 earthquake. The energy released as seismic waves increases even more with earthquake magnitude, by a factor of about 32 for each magnitude unit. A magnitude 7 earthquake releases 32×32 , or about 1000, times the energy of a magnitude 5 earthquake. According to this energy scale, the Tohoku megaquake was a million times more powerful than a magnitude 5 event!

Seismic waves gradually weaken as they move away from the focus, so to make his procedure work for any seismograph, Richter had to find a way to correct the

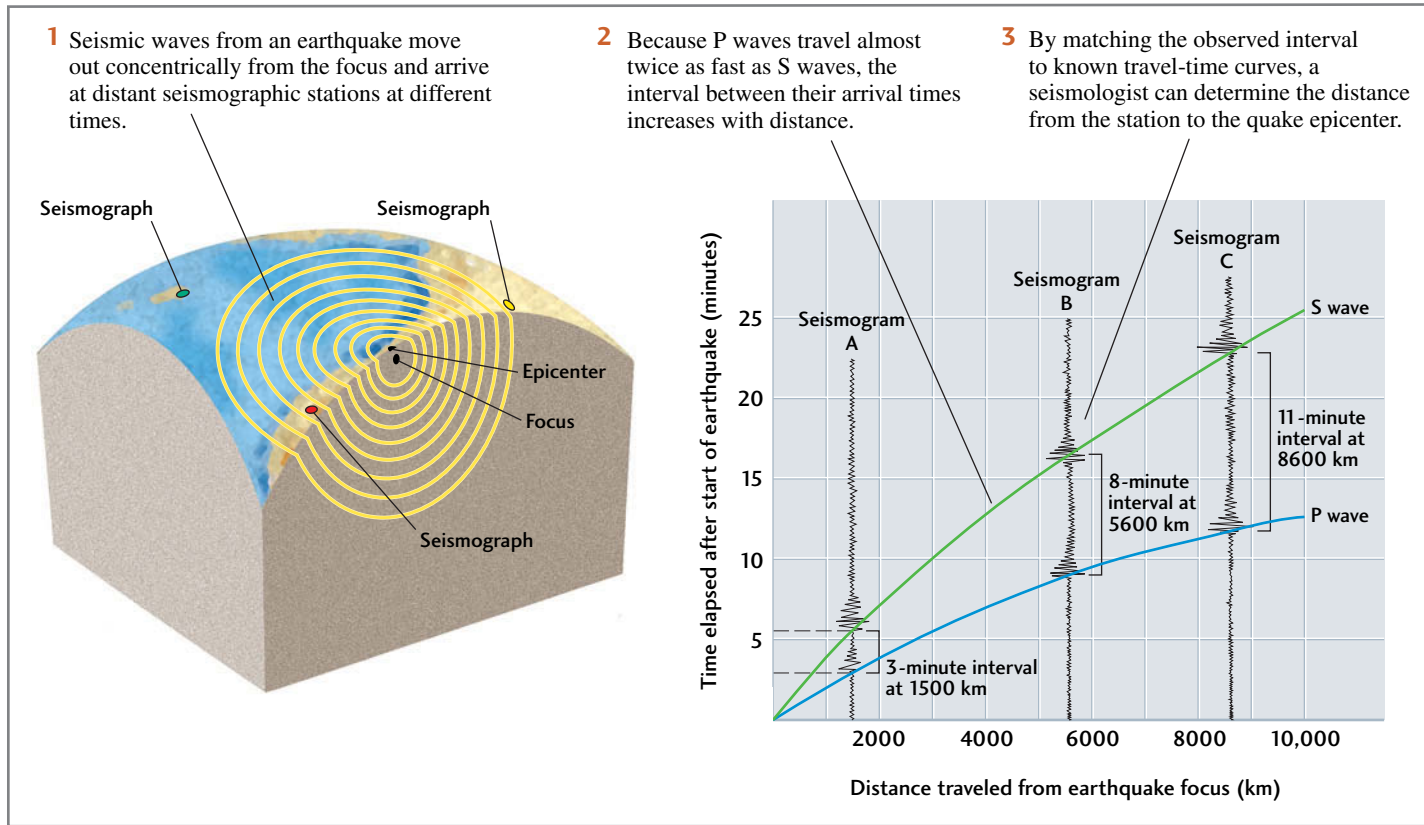


FIGURE 13.10 ■ Readings from three or more seismographic stations can be used to determine the location of an earthquake's focus.

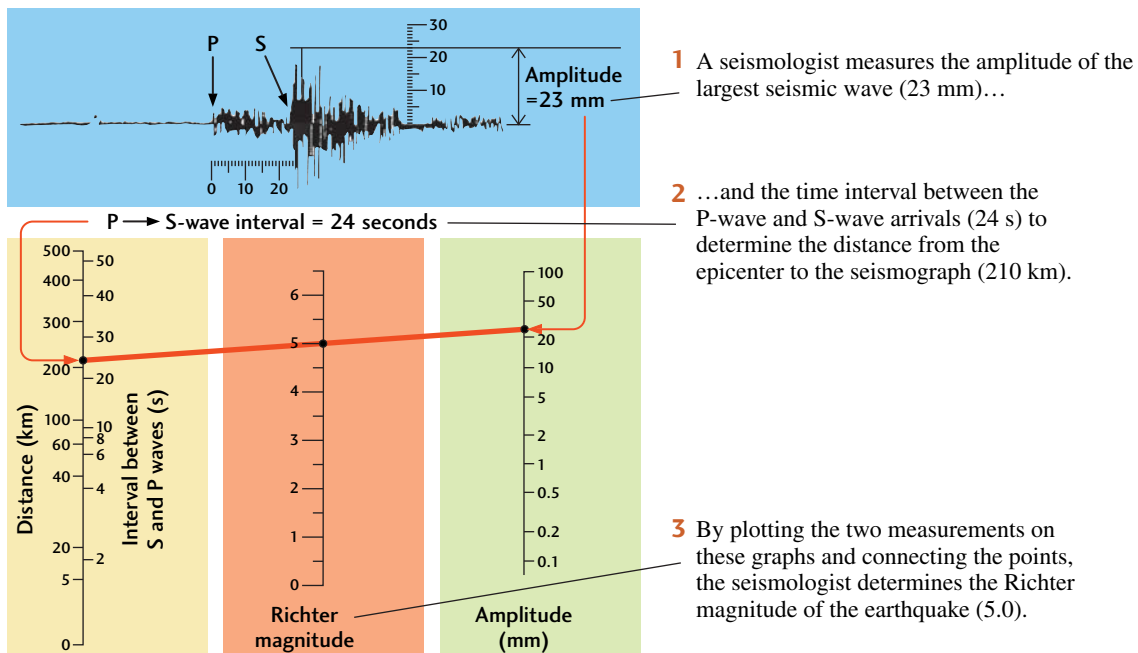
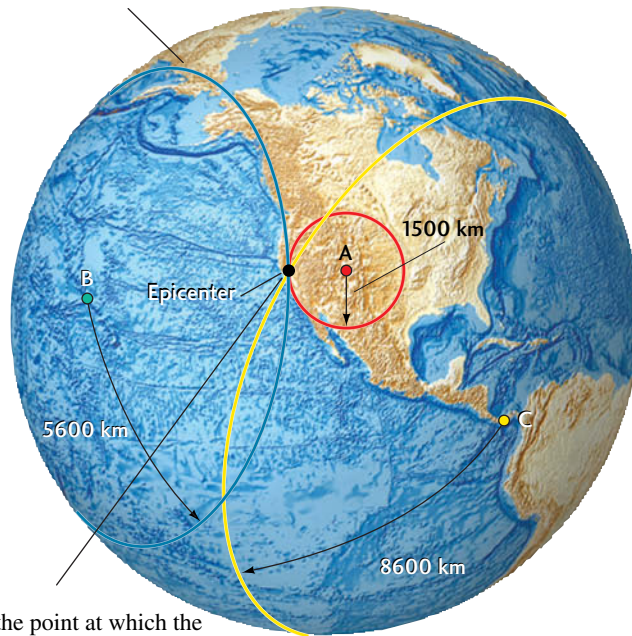


FIGURE 13.11 ■ The maximum amplitude of ground movement, corrected by the P-S wave interval, is used to assign a Richter magnitude to an earthquake. [California Institute of Technology.]

- 4 If the seismologist then draws a circle with a radius calculated from the travel-time curves around each seismographic station,...



- 5 ...the point at which the circles intersect will locate the earthquake's epicenter.

measurement of ground movement for the distance between the seismograph and the focus. He devised a simple graph that allowed seismologists at different locations to quickly come up with nearly the same value for the magnitude of an earthquake no matter how far their instruments were from the focus (see Figure 13.11). His procedure came to be used throughout the world.

MOMENT MAGNITUDE Although "Richter scale" has become a household term, seismologists prefer a measure of earthquake size more directly related to the physical properties of the faulting that causes an earthquake. The *seismic moment* of an earthquake is defined as a number proportional to the product of the area of faulting and the average fault slip. The corresponding *moment magnitude* increases by about one unit for every tenfold increase in the area of faulting (see the Practicing Geology exercise at the end of the chapter).

Although Richter's method and the moment method produce roughly the same numerical values, the moment magnitude can be measured more accurately from seismographs, and it can sometimes be determined directly from field measurements of the fault.

MAGNITUDE AND FREQUENCY Large earthquakes occur much less often than small ones. This observation can be expressed by a simple relationship between earthquake frequency and magnitude (Figure 13.12). Worldwide,

Magnitude

Energy release (equivalent kilograms of explosive)

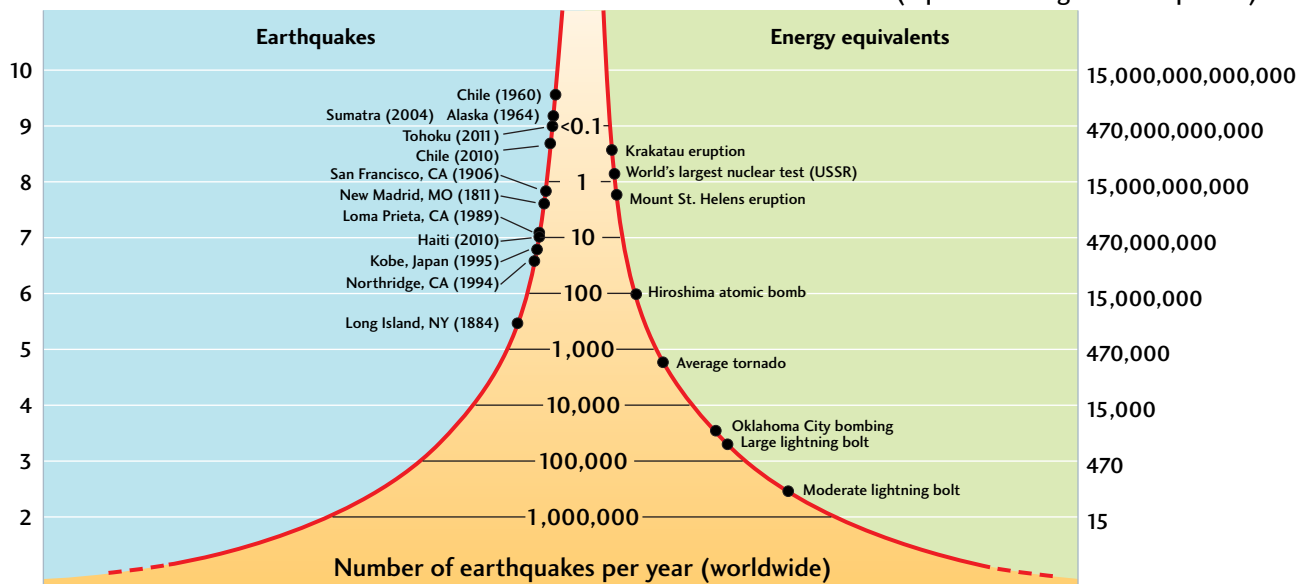


FIGURE 13.12 Relationship between moment magnitude, seismic energy release, and number of earthquakes per year worldwide. Examples of earthquakes of various magnitudes and of other large sources of sudden energy release are included for comparison. [Adapted from IRIS Consortium, <http://www.iris.edu>.]

TABLE 13-1 Modified Mercalli Intensity Scale

Intensity Level	Description
I	Not felt.
II	Felt only by a few people at rest. Suspended objects may swing.
III	Felt noticeably indoors. Many people do not recognize it as an earthquake. Parked cars may rock slightly.
IV	Felt indoors by many, outdoors by few. Dishes, windows, doors rattle. Parked cars rock noticeably.
V	Felt by most; many awakened. Some dishes, windows broken. Unstable objects overturned.
VI	Felt by all. Some heavy furniture moves. Damage slight.
VII	Slight to moderate damage in well-built structures; considerable damage in poorly built structures; some chimneys broken.
VIII	Considerable damage in well-built structures. Damage great in poorly built structures. Fall of chimneys, factory stacks, columns, monuments, walls.
IX	Damage great in well-built structures, with partial collapse. Buildings shifted off foundations.
X	Some well-built wooden structures destroyed; most masonry and frame structures destroyed. Rails bent.
XI	Few if any masonry structures remain standing. Bridges destroyed. Rails bent greatly.
XII	Damage total. Lines of sight and level are distorted. Objects thrown into the air.

approximately 1,000,000 earthquakes with magnitudes greater than 2 take place each year. This number decreases by a factor of 10 for each magnitude unit. Hence, there are about 100,000 earthquakes with magnitudes greater than 3, about 1000 with magnitudes greater than 5, and about 10 with magnitudes greater than 7 each year.

According to these statistics, there should be, on average, about 1 earthquake with a magnitude greater than 8 per year and 1 earthquake with a magnitude greater than 9 every 10 years. In fact, the very largest earthquakes, such as the ones that occurred on thrust faults in the subduction zones off Japan in 2011 (moment magnitude 9.0), Sumatra in 2004 (moment magnitude 9.2), Alaska in 1964 (moment magnitude 9.1), and Chile in 1960 (moment magnitude 9.5) and 2010 (moment magnitude 8.8) are almost this common when averaged over many decades. However, even the largest subduction zone megathrusts are too small to create magnitude 10 earthquakes, so seismologists believe that events of such extreme size do not follow this rule; that is, they are less frequent than once per century.

SHAKING INTENSITY Earthquake magnitude by itself does not describe seismic hazard because the shaking that causes destruction generally weakens with distance from the fault rupture. A magnitude 8 earthquake in a remote area far from the nearest city might cause no human or economic losses, whereas a magnitude 6 quake immediately beneath a city would likely cause serious damage. The destruction in

Christchurch by the earthquakes of February 22 and June 13, 2011, illustrates this important point (see Figure 13.7).

In the late nineteenth century, before Richter invented his magnitude scale, seismologists and earthquake engineers developed **intensity scales** for estimating the intensity of shaking directly from an earthquake's destructive effects. **Table 13.1** shows one intensity scale that remains in common use today, called the *modified Mercalli intensity scale* after Giuseppe Mercalli, the Italian scientist who proposed it in 1902. This scale assigns a value, given as a Roman numeral from I to XII, to the intensity of the shaking at a particular location. For example, a location where an earthquake is barely felt by a few people is assigned an intensity of II, whereas one where it was felt by nearly everyone is assigned an intensity of V. Numbers at the upper end of the scale indicate increasing amounts of damage. The narrative attached to the highest value, XII, is tersely apocalyptic: "Damage total. Lines of sight and level are distorted. Objects thrown into the air."

By making observations at many sites and interviewing many people who experienced an earthquake, or even by examining historical records, seismologists can make maps showing contours of equal shaking intensity. **Figure 13.13** shows an intensity map for the New Madrid earthquake of December 16, 1811, a magnitude 7.7 event near the southern tip of Missouri, which was felt as far away as Boston. Although earthquake intensities are generally highest near the fault rupture, they also depend on the local geology.

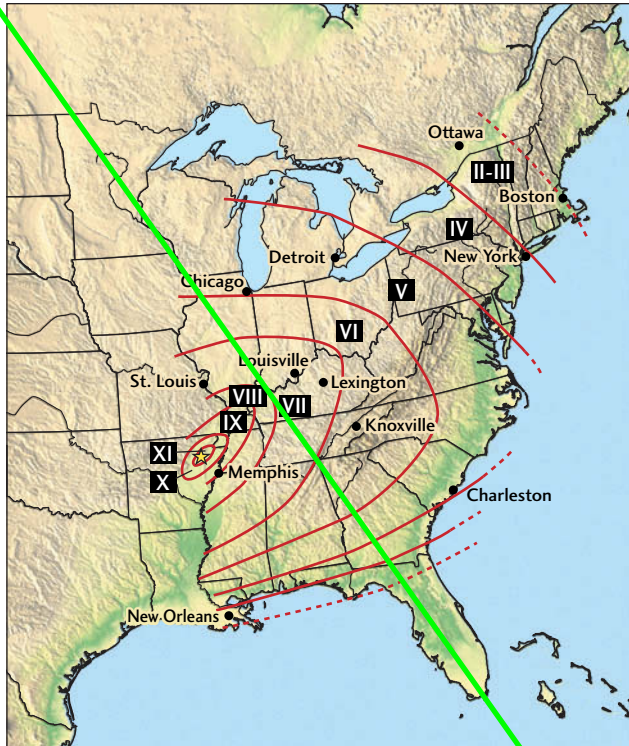


FIGURE 13.13 ■ Measurements of modified Mercalli intensities for the New Madrid earthquake of December 16, 1811, a magnitude 7.7 event near the juncture of Missouri, Arkansas, and Tennessee. Regions near the epicenter experienced intensities greater than IX, and intensities as high as VI were observed 200 km from the epicenter (see Table 13.1). [Carl W. Stover and Jerry L. Cossman, USGS Professional Paper 1527, 1993.]

For example, when sites at equal distances from the rupture are compared, the shaking tends to be more intense on soft sediments (especially water-saturated sediments near shorelines) than on hard basement rock. Intensity maps thus provide engineers with crucial data for designing structures to withstand seismic shaking.

Determining Fault Mechanisms

The pattern of ground shaking also depends on the orientation of the fault rupture and the direction of slipping, which together specify the **fault mechanism** of an earthquake. The fault mechanism tells us whether the rupture was on a *normal*, *reverse*, or *strike-slip* fault. If the rupture was on a strike-slip fault, the fault mechanism also tells us whether the movement was *right-lateral* or *left-lateral* (see Figure 7.8 for the definitions of these terms). We can then use this information to infer the regional pattern of tectonic forces (**Figure 13.14**).

For shallow ruptures that break the surface, we can sometimes determine the fault mechanism from field observations of the fault scarp. As we have seen, however, most ruptures are too deep to break the surface, so we must deduce the fault mechanism from seismograms.

For large earthquakes at any depth, this task turns out to be easy, because there are enough seismographic stations around the world to surround any earthquake's focus. In some directions from the focus, the very first movement of the ground recorded by a seismograph—a P wave—is a *push away* from the focus, causing upward movement on a vertical seismograph. In other directions, the initial ground movement is a *pull toward* the focus, causing downward movement on a vertical seismograph. For strike-slip ruptures, the directions of the largest pushes lie on an axis rotated 45° from the fault plane and are perpendicular to the directions of the largest pulls (**Figure 13.15**). The locations of pushes and pulls can therefore be plotted and divided into four sections based on the positions of the seismographic stations. One of the two boundaries between those sections will be the fault orientation; the other will be a plane perpendicular to the fault. The slip direction on the fault plane can also be determined from the arrangement of pushes and pulls. In this manner, without surface evidence, seismologists can deduce whether the horizontal crustal forces that triggered an earthquake were tensional, compressive, or shearing forces.

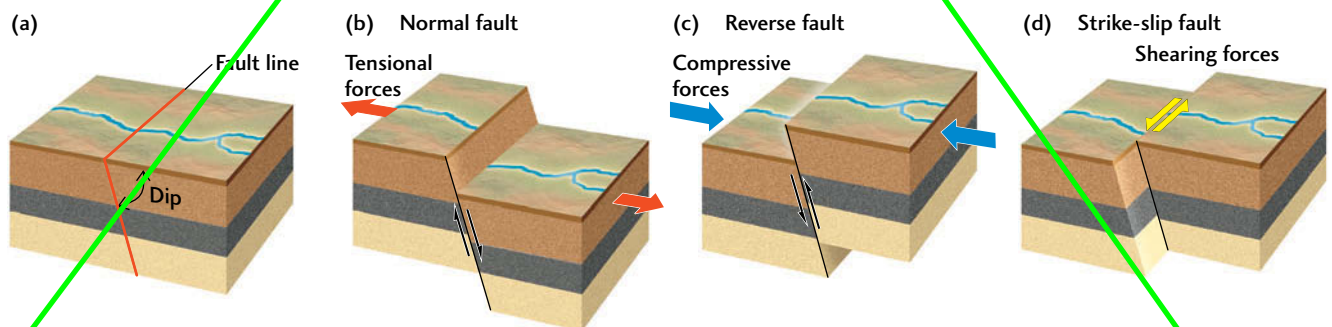


FIGURE 13.14 ■ The three main types of fault mechanisms that initiate earthquakes and the stresses that cause them. (a) A fault before movement takes place. (b) Normal faulting due to tensional stress. (c) Reverse faulting due to compressive stress. (d) Strike-slip faulting due to shearing stress (in this case, left-lateral).

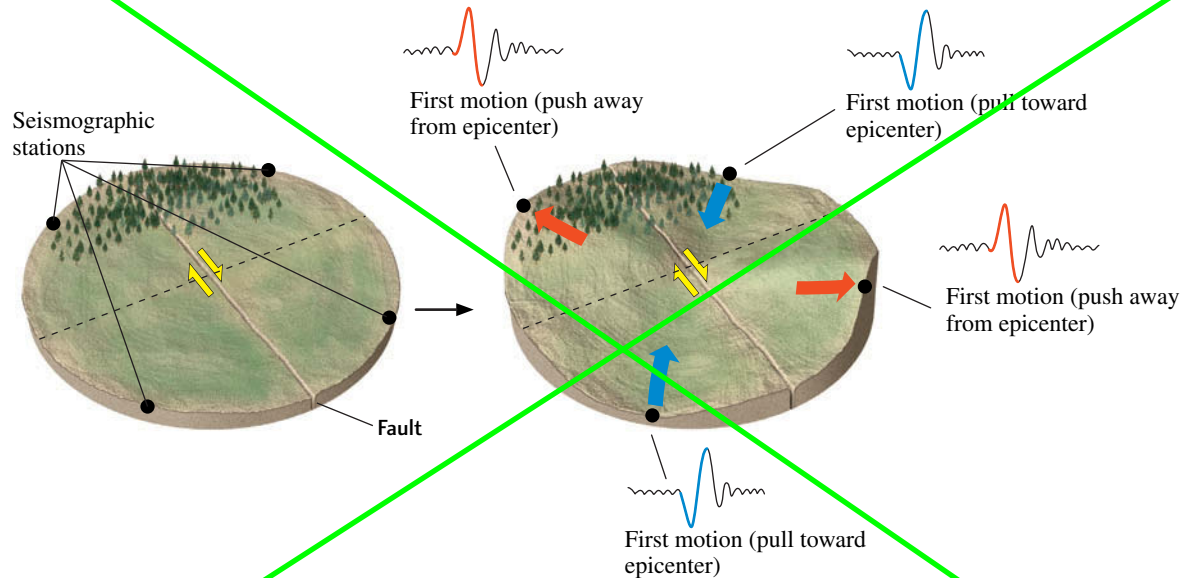


FIGURE 13.15 ■ The movement marking the first P wave arriving from a fault rupture at each of several seismographic stations is used to determine the orientation of the fault and the direction of fault slipping. The case shown here is the rupture of a right-lateral strike-slip fault. Note that the alternating pattern of pushes and pulls would remain the same if a plane perpendicular to the fault ruptured with left-lateral displacement. Seismologists can usually choose between the two possibilities using additional information, such as field mapping of the fault scarp or the alignment of aftershocks along the fault.

GPS Measurements and “Silent” Earthquakes

As we saw in Chapter 2, GPS receivers can record the slow movements of lithospheric plates. These instruments can also measure the strain that builds up from such movements, as well as the sudden slipping on a fault when it ruptures in an earthquake.

Seismologists use GPS observations to study another kind of movement along active faults. It has been known for many years that a section of the San Andreas fault in central California creeps continuously, rather than rupturing suddenly. This creep slowly deforms structures and cracks pavements that cross the fault trace. More recently, GPS receivers have found surface movements at convergent plate boundaries that reflect short-lived creep events, which commonly last days to weeks at a time. They have been named *silent earthquakes* because these gradual movements do not trigger destructive seismic waves. Nevertheless, they can release large amounts of stored strain energy.

These observations raise many questions that geologists are now trying to answer: What causes faults to stick and slip catastrophically in some places and creep in others? Will the release of strain energy by silent earthquakes make destructive earthquakes in those regions less frequent or less severe? Can silent earthquakes be used to predict potentially destructive earthquakes?

Earthquakes and Patterns of Faulting

As we have seen, seismologists are using networks of sensitive seismographs to locate earthquakes around the world, measure their magnitudes, and deduce their fault mechanisms. These methods are revealing new information about tectonic processes on scales much smaller than the plates themselves. In this section, we summarize the global pattern of earthquake occurrence from the perspective of plate tectonics and show how regional studies of active fault systems are improving our understanding of earthquakes along plate boundaries and within plate interiors.

The Big Picture: Earthquakes and Plate Tectonics

The *seismicity map* in **Figure 13.16** shows the epicenters of earthquakes recorded around the world since 1976. The most obvious features of this map, known to seismologists for many decades, are the belts of seismic activity that mark the major plate boundaries. The fault mechanisms observed for earthquakes in these belts (**Figure 13.17**) are consistent with the types of faulting along different types of plate boundaries that we described in Chapter 7.

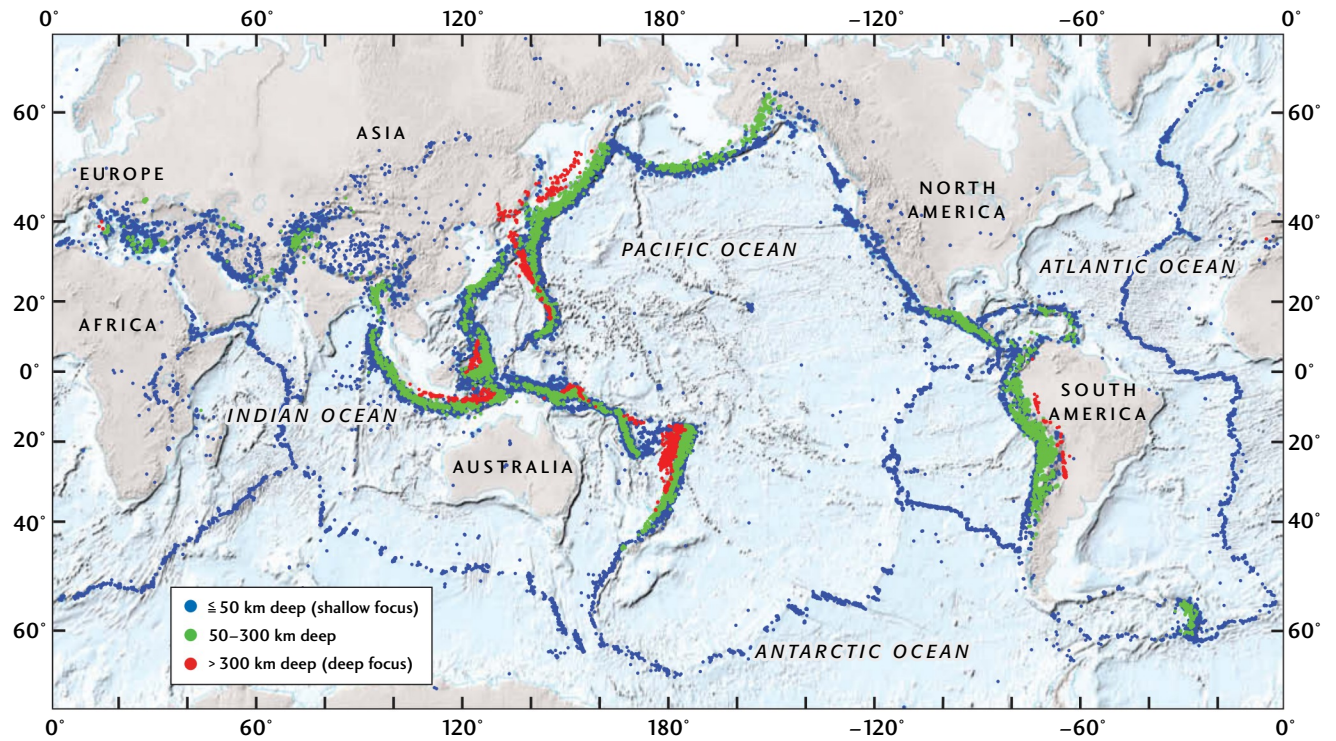


FIGURE 13.16 ■ Global map of seismic activity from January, 1976, through October, 2013. Each dot represents the epicenter of an earthquake larger than magnitude 5. Colors indicate the focal depth. Note the concentration of earthquakes along the boundaries between major lithospheric plates. [Map based on data from Global CMT catalog; plot by M. Boettcher.]

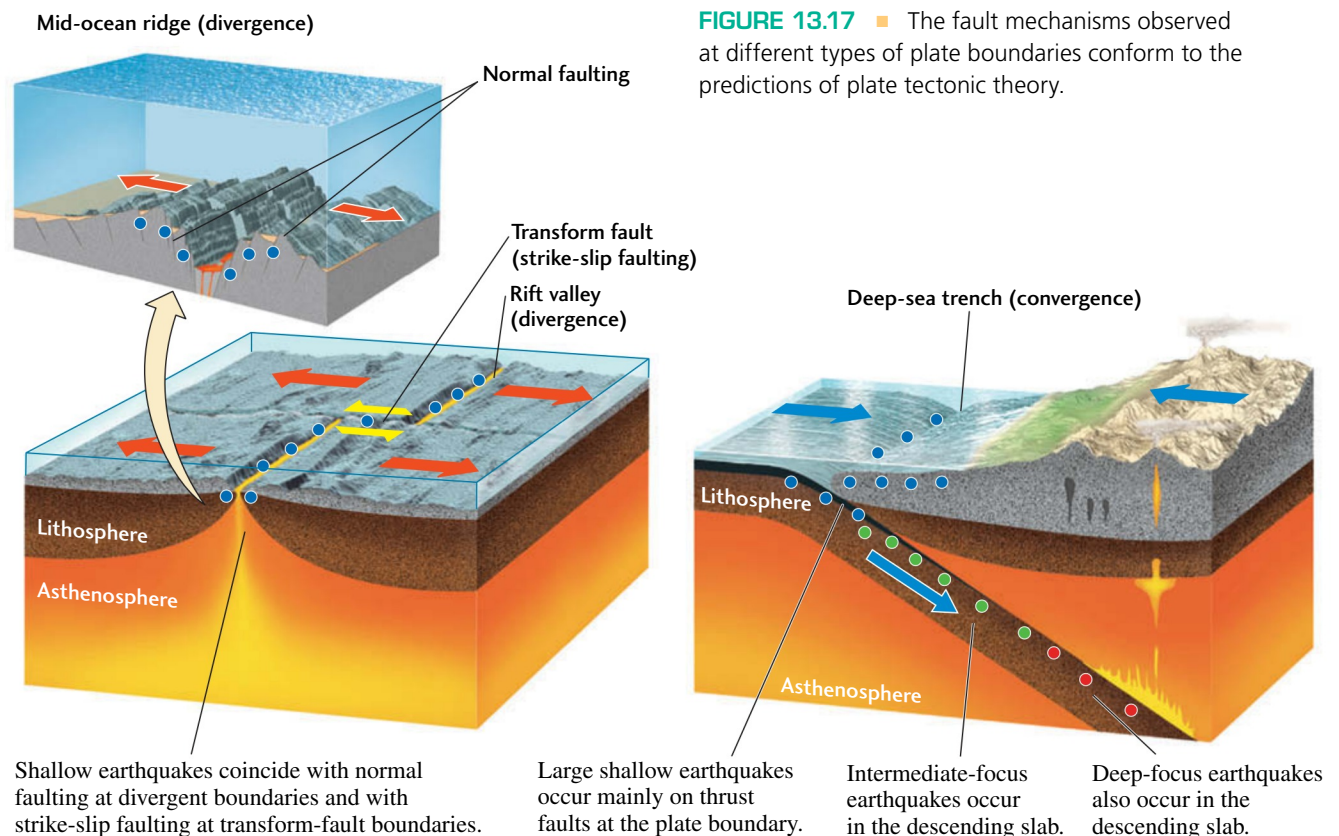


FIGURE 13.17 ■ The fault mechanisms observed at different types of plate boundaries conform to the predictions of plate tectonic theory.

DIVERGENT BOUNDARIES The narrow belts of shallow earthquakes that run through ocean basins coincide with mid-ocean ridge crests and their offsets on transform faults. The P waves recorded from the ridge-crest quakes indicate that they are caused by normal faulting. The faults strike parallel to the ridge and dip toward the rift valley at the ridge crest. Normal faulting implies that tensional forces are at work as the plates are pulled apart during seafloor spreading. Earthquakes also have normal fault mechanisms in zones where continental crust is being pulled apart, such as in the East African rift valleys and in the Basin and Range province of western North America.

TRANSFORM-FAULT BOUNDARIES Earthquake activity is even greater along the transform-fault boundaries that offset mid-ocean ridge segments. These earthquakes have strike-slip fault mechanisms, just as one would expect where plates create horizontal shearing forces as they slide past each other in opposite directions. Moreover, for earthquakes along these transform faults, the slip direction indicated by the fault mechanisms is left-lateral where the ridge crest steps right and right-lateral where it steps left. These directions are the opposite of what would be needed to create the offsets of the ridge crest, but are consistent with the direction of slip predicted by seafloor spreading. In the mid-1960s, seismologists used this property of the transform faults to support the hypothesis of seafloor spreading. Slip directions on transform faults that run through continental crust, such as California's San Andreas fault and New Zealand's Alpine fault (both right-lateral), also agree with the predictions of plate tectonics.

CONVERGENT BOUNDARIES The world's largest earthquakes occur at convergent plate boundaries. The four greatest earthquakes of the last hundred years were of this type: the 2011 Tohoku earthquake (magnitude 9.0); the Sumatra earthquake of 2004; the Alaska earthquake of 1964; and the largest of all, the 1960 earthquake in the subduction zone west of Chile. During the Chile earthquake, the crust of the Nazca Plate slipped an average of 20 m beneath the crust of the South American Plate on a fault rupture with an area larger than Arizona! The fault mechanisms of these great earthquakes show that they were caused by horizontal compression along *megathrusts*, the huge thrust faults that form the boundaries where one plate is subducted beneath another. All three of these earthquakes displaced the seafloor, generating tsunamis that devastated coastlines.

Earth's deepest earthquakes also occur at convergent boundaries. Almost all earthquakes originating below 100 km rupture the descending plate in a subduction zone. The fault mechanisms of these deep earthquakes show a variety of orientations, but they are consistent with the deformation expected within the descending plate as gravity pulls it back into the convecting mantle. The deepest earthquakes take place in the oldest—and therefore

coldest—descending plates, such as those beneath South America, Japan, and the island arcs of the western Pacific Ocean.

INTRAPLATE EARTHQUAKES Although most earthquakes occur at plate boundaries, a small percentage of global seismic activity originates within plate interiors. The foci of these *intraplate earthquakes* are relatively shallow, and most occur on continents. Among these earthquakes are some of the most famous in American history: a sequence of three large events near New Madrid, Missouri, in 1811–1812; the Charleston, South Carolina, earthquake of 1886; and the Cape Ann earthquake, near Boston, Massachusetts, in 1755. Many intraplate earthquakes occur on old faults that were once parts of ancient plate boundaries. These faults no longer form plate boundaries, but they remain zones of crustal weakness that concentrate and release intraplate stresses.

One of the deadliest intraplate earthquakes (magnitude 7.6) occurred near Bhuj, in the state of Gujarat, in western India, in 2001. It is estimated that some 20,000 lives were lost. The Bhuj earthquake occurred on a previously unknown thrust fault 1000 km south of the boundary between the Indian and Eurasian plates, but the compressive stresses responsible for this rupture resulted from the ongoing collision of these two plates. Intraplate earthquakes show that strong crustal forces can develop and cause faulting within a lithospheric plate far from modern plate boundaries.

Regional Fault Systems

Although the fault mechanisms of most major earthquakes conform to the predictions of plate tectonic theory, a plate boundary can rarely be described as a single fault, particularly when the boundary involves continental crust. Rather, the zone of deformation between two moving plates usually comprises a network of interacting faults—a *fault system*. The fault system in Southern California provides an interesting example (Figure 13.18).

The “master fault” of this system is our old nemesis, the San Andreas—a right-lateral strike-slip fault that runs northwestward through California from the Salton Sea near the Mexican border until it goes offshore in the northern part of the state (see Figure 7.7). There are a number of subsidiary faults on either side of the San Andreas, however, that generate large earthquakes. In fact, most of the damaging earthquakes in Southern California during the past century have occurred on those subsidiary faults.

Why is the San Andreas fault system so complex? Part of the explanation has to do with the geometry of the San Andreas fault itself. A bend in the fault creates compressive forces that cause thrust faulting in the area north of Los Angeles (see Figure 7.22). Thrust faulting in this “Big Bend” was responsible for two recent deadly earthquakes, the San Fernando earthquake of 1971 (magnitude 6.6, 65 people killed) and the Northridge earthquake of 1994 (magnitude 6.7,